

Project #47: LogLens

AI Portfolio Project Report

Project Overview

Project: #47 - LogLens

Language: Perl 5.34.1

Framework: Mojolicious 9.42

Lines of Code: 906

Date: 2026-03-29

GitHub: <https://github.com/malbinjvc/log-lens>

LogLens is an AI-powered log analysis REST API built with Perl and Mojolicious. It provides intelligent log parsing, pattern detection, anomaly identification, and AI-driven insights for application logs. Features a mock Claude AI client for generating analysis summaries.

Architecture

- Mojolicious web framework with non-blocking I/O
- In-memory data store for log entries
- Service layer: LogLens::Services for business logic
- Client layer: LogLens::Clients for AI integration
- Model layer: LogLens::Models for data structures
- RESTful JSON API on port 8080

Key Features

- Log entry creation with severity levels (DEBUG, INFO, WARN, ERROR, FATAL)
- Pattern detection across log entries using regex matching
- Anomaly detection with configurable thresholds
- AI-powered log analysis via mock Claude client
- Log statistics and severity distribution
- Log filtering by severity, time range, and source
- Bulk log import and export functionality

API Endpoints

- GET /health - Health check
- POST /api/logs - Create log entry
- GET /api/logs - List/filter log entries
- GET /api/logs/:id - Get log entry by ID
- POST /api/logs/analyze - AI-powered log analysis
- GET /api/logs/patterns - Detect patterns in logs
- GET /api/logs/anomalies - Detect anomalies
- GET /api/logs/stats - Log statistics

Testing

Test Results: 183/183 passing

Errors Encountered: 3

Errors Resolved: 3

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- PERL5LIB path required for Mojolicious module resolution
- Uninitialized \$_ warning in pattern matching (non-fatal)
- Docker Perl module installation via cpanm in build stage

Docker

- Multi-stage build with perl:5.40-slim
- Non-root user (loglens)
- Healthcheck via Mojo::UserAgent against /health
- Runtime image includes only Perl and required modules

CI/CD Pipeline

- GitHub Actions on ubuntu-latest
- Perl 5 with cpanm for dependencies
- prove -l t/ for test execution
- Docker build verification step
- All jobs passing

Final Status

Build: Passing

Tests: 183/183 green

CI: All green

Docker: Build successful

GitHub: Pushed and live